

COM3110/4115:

Text Processing

Introduction

Rob Gaizauskas

Department of Computer Science
University of Sheffield

Lecturers Rob Gaizauskas (r.gaizauskas@sheffield.ac.uk)
 Carolina Scarton (c.scarton@sheffield.ac.uk)

- All materials for COM 3110 / 4115 available on Blackboard including:
 - ◇ key course details
 - ◇ lecture slides + Encore recordings of the lectures
 - ◇ lab sheets and details of the Python setup needed for the labs
 - ◇ assignments
 - ◇ links to past exam papers
 - ◇ announcements
 - ◇ discussion boards/forums

Course Aims

- To develop an understanding of the fundamentals of how digitally stored text is represented and processed in a computer.

Course Aims

- To develop an understanding of the fundamentals of how digitally stored text is represented and processed in a computer.
- Develop ability to construct simple systems for applying such techniques.

Course Aims

- To develop an understanding of the fundamentals of how digitally stored text is represented and processed in a computer.
- Develop ability to construct simple systems for applying such techniques.
- To develop an understanding of the basic problems and principles underlying text processing applications.

Course Aims

- To develop an understanding of the fundamentals of how digitally stored text is represented and processed in a computer.
- Develop ability to construct simple systems for applying such techniques.
- To develop an understanding of the basic problems and principles underlying text processing applications.
- (COM4115 only) For one or more topics in text processing to explore refinements beyond the most basic approaches.

Course Aims

- To develop an understanding of the fundamentals of how digitally stored text is represented and processed in a computer.
- Develop ability to construct simple systems for applying such techniques.
- To develop an understanding of the basic problems and principles underlying text processing applications.
- (COM4115 only) For one or more topics in text processing to explore refinements beyond the most basic approaches.

Prerequisites:

Course Aims

- To develop an understanding of the fundamentals of how digitally stored text is represented and processed in a computer.
- Develop ability to construct simple systems for applying such techniques.
- To develop an understanding of the basic problems and principles underlying text processing applications.
- (COM4115 only) For one or more topics in text processing to explore refinements beyond the most basic approaches.

Prerequisites:

- Interest in language and basic knowledge of English

Course Aims

- To develop an understanding of the fundamentals of how digitally stored text is represented and processed in a computer.
- Develop ability to construct simple systems for applying such techniques.
- To develop an understanding of the basic problems and principles underlying text processing applications.
- (COM4115 only) For one or more topics in text processing to explore refinements beyond the most basic approaches.

Prerequisites:

- Interest in language and basic knowledge of English
- Some mathematical basics, e.g. basic probability theory

Course Aims

- To develop an understanding of the fundamentals of how digitally stored text is represented and processed in a computer.
- Develop ability to construct simple systems for applying such techniques.
- To develop an understanding of the basic problems and principles underlying text processing applications.
- (COM4115 only) For one or more topics in text processing to explore refinements beyond the most basic approaches.

Prerequisites:

- Interest in language and basic knowledge of English
- Some mathematical basics, e.g. basic probability theory
- **Some programming skills including basic familiarity with Python**

By the end of the module, you should be able to:

- code in a programming language well-suited to text handling (Python)

By the end of the module, you should be able to:

- code in a programming language well-suited to text handling (Python)
- identify and explain key techniques that are relevant to performing a number of text processing tasks

By the end of the module, you should be able to:

- code in a programming language well-suited to text handling (Python)
- identify and explain key techniques that are relevant to performing a number of text processing tasks
- implement systems able to analyse large volumes of textual data, and to perform basic text processing tasks and (for COM4115), in selected cases, more advanced text processing tasks

Motivation

What is text processing and why study it? Proposed definition:

Motivation

What is text processing and why study it? Proposed definition:

*The creation, storage and access of text in digital form
by computer*

Motivation

What is text processing and why study it? Proposed definition:

*The creation, storage and access of text in digital form
by computer*

Reasons for studying text processing now include:

Motivation

What is text processing and why study it? Proposed definition:

*The creation, storage and access of text in digital form
by computer*

Reasons for studying text processing now include:

- **Digital Text Collections/Streams:** The Web, Corporate Intranets, Social Media, Digital Archives

Motivation

What is text processing and why study it? Proposed definition:

The creation, storage and access of text in digital form by computer

Reasons for studying text processing now include:

- **Digital Text Collections/Streams:** The Web, Corporate Intranets, Social Media, Digital Archives
 - ◇ Access – more text than ever, available to more people than ever, in more languages than ever

Motivation

What is text processing and why study it? Proposed definition:

The creation, storage and access of text in digital form by computer

Reasons for studying text processing now include:

- **Digital Text Collections/Streams:** The Web, Corporate Intranets, Social Media, Digital Archives
 - ◇ Access – more text than ever, available to more people than ever, in more languages than ever
 - widely discussed problem: *information overload*

Motivation

What is text processing and why study it? Proposed definition:

The creation, storage and access of text in digital form by computer

Reasons for studying text processing now include:

- **Digital Text Collections/Streams:** The Web, Corporate Intranets, Social Media, Digital Archives
 - ◇ Access – more text than ever, available to more people than ever, in more languages than ever
 - widely discussed problem: *information overload*
 - premium on technology that can facilitate *information access and insight*

Motivation

What is text processing and why study it? Proposed definition:

The creation, storage and access of text in digital form by computer

Reasons for studying text processing now include:

- **Digital Text Collections/Streams:** The Web, Corporate Intranets, Social Media, Digital Archives
 - ◇ Access – more text than ever, available to more people than ever, in more languages than ever
 - widely discussed problem: *information overload*
 - premium on technology that can facilitate *information access* and *insight*
 - ◇ *Creation* – automatic creation/update of web content, textual expression of sensor data/structured data

Motivation (contd)

... reasons for studying text processing (contd):

Motivation (contd)

... reasons for studying text processing (contd):

- **Convergence with NLP**

... reasons for studying text processing (contd):

- **Convergence with NLP**

- ◊ *NLP* (natural language processing) seeks to build programs that can “understand” texts

... reasons for studying text processing (contd):

- **Convergence with NLP**

- ◇ *NLP* (natural language processing) seeks to build programs that can “understand” texts
- ◇ *Text Processing* – usually seen to have more modest, engineering aims

... reasons for studying text processing (contd):

- **Convergence with NLP**

- ◇ *NLP* (natural language processing) seeks to build programs that can “understand” texts
- ◇ *Text Processing* – usually seen to have more modest, engineering aims
- ◇ *Convergence* – increasingly they are borrowing ideas and techniques from each other

... reasons for studying text processing (contd):

- **Convergence with NLP**

- ◇ *NLP* (natural language processing) seeks to build programs that can “understand” texts
- ◇ *Text Processing* – usually seen to have more modest, engineering aims
- ◇ *Convergence* – increasingly they are borrowing ideas and techniques from each other
 - particularly in area of *statistical/neural language processing*

... reasons for studying text processing (contd):

- **Convergence with NLP**

- ◇ *NLP* (natural language processing) seeks to build programs that can “understand” texts
- ◇ *Text Processing* – usually seen to have more modest, engineering aims
- ◇ *Convergence* – increasingly they are borrowing ideas and techniques from each other
 - particularly in area of *statistical/neural language processing*
- ◇ Large language models (LLMs), such as that underlying ChatGPT now at the forefront of AI

Applications: Text Processing or NLP?

Distinction commonly seen in terms of whether task requires some 'understanding' of language, or special linguistic knowledge.

Some common text processing/NLP applications:

- Information Retrieval

Applications: Text Processing or NLP?

Distinction commonly seen in terms of whether task requires some 'understanding' of language, or special linguistic knowledge.

Some common text processing/NLP applications:

- Information Retrieval
- Information Extraction/Text Mining

Applications: Text Processing or NLP?

Distinction commonly seen in terms of whether task requires some 'understanding' of language, or special linguistic knowledge.

Some common text processing/NLP applications:

- Information Retrieval
- Information Extraction/Text Mining
- Text Categorisation

Applications: Text Processing or NLP?

Distinction commonly seen in terms of whether task requires some 'understanding' of language, or special linguistic knowledge.

Some common text processing/NLP applications:

- Information Retrieval
- Information Extraction/Text Mining
- Text Categorisation
- Automatic Summarisation

Applications: Text Processing or NLP?

Distinction commonly seen in terms of whether task requires some 'understanding' of language, or special linguistic knowledge.

Some common text processing/NLP applications:

- Information Retrieval
- Information Extraction/Text Mining
- Text Categorisation
- Automatic Summarisation
- NL Generation

Applications: Text Processing or NLP?

Distinction commonly seen in terms of whether task requires some 'understanding' of language, or special linguistic knowledge.

Some common text processing/NLP applications:

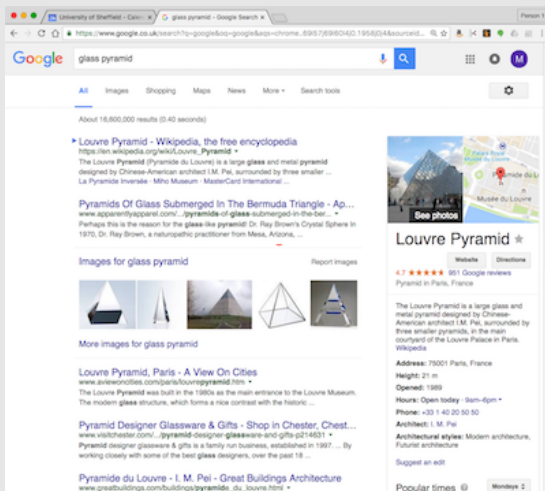
- Information Retrieval
- Information Extraction/Text Mining
- Text Categorisation
- Automatic Summarisation
- NL Generation
- Machine Translation

Applications: IR

Information Retrieval (IR): concerned with developing algorithms and models for retrieving *relevant* documents from text collections.

Applications: IR

Information Retrieval (IR): concerned with developing algorithms and models for retrieving *relevant* documents from text collections.



Applications: IR (contd)

- Text collection = some set of 'documents'

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◇ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◇ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts
 - ◇ now, billions of pages on the WWW

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◊ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts
 - ◊ now, billions of pages on the WWW
- Query: user indication of what s/he wants

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◊ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts
 - ◊ now, billions of pages on the WWW
- Query: user indication of what s/he wants
 - ◊ commonly, just 2 or 3 words — good basis for retrieval?

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◇ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts
 - ◇ now, billions of pages on the WWW
- Query: user indication of what s/he wants
 - ◇ commonly, just 2 or 3 words — good basis for retrieval?
- How decide what docs are relevant?

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◇ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts
 - ◇ now, billions of pages on the WWW
- Query: user indication of what s/he wants
 - ◇ commonly, just 2 or 3 words — good basis for retrieval?
- How decide what docs are relevant?
 - ◇ how decide if one method works better than another?

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◊ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts
 - ◊ now, billions of pages on the WWW
- Query: user indication of what s/he wants
 - ◊ commonly, just 2 or 3 words — good basis for retrieval?
- How decide what docs are relevant?
 - ◊ how decide if one method works better than another?
- Much work is still left to the user:

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◇ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts
 - ◇ now, billions of pages on the WWW
- Query: user indication of what s/he wants
 - ◇ commonly, just 2 or 3 words — good basis for retrieval?
- How decide what docs are relevant?
 - ◇ how decide if one method works better than another?
- Much work is still left to the user:
 - ◇ task of selecting which of returned docs *are* relevant

Applications: IR (contd)

- Text collection = some set of 'documents'
 - ◇ originally, few hundred/thousand electronically stored documents, e.g. journal paper abstracts
 - ◇ now, billions of pages on the WWW
- Query: user indication of what s/he wants
 - ◇ commonly, just 2 or 3 words — good basis for retrieval?
- How decide what docs are relevant?
 - ◇ how decide if one method works better than another?
- Much work is still left to the user:
 - ◇ task of selecting which of returned docs *are* relevant
 - ◇ task of *extracting* the relevant information

- IR contrasts with *Information Extraction* (IE).

Applications: IE

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes

Applications: IE

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes
 - ◇ type information to be extracted must be decided *in advance*

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes
 - ◇ type information to be extracted must be decided *in advance*
 - ◇ information extracted typically includes mentions of

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes
 - ◇ type information to be extracted must be decided *in advance*
 - ◇ information extracted typically includes mentions of
 - *entities* (e.g. organisations, persons, locations) and

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes
 - ◇ type information to be extracted must be decided *in advance*
 - ◇ information extracted typically includes mentions of
 - *entities* (e.g. organisations, persons, locations) and
 - *relations* (e.g. *person* IS-EMPLOYED-BY *organisation*)

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes
 - ◇ type information to be extracted must be decided *in advance*
 - ◇ information extracted typically includes mentions of
 - *entities* (e.g. organisations, persons, locations) and
 - *relations* (e.g. *person* IS-EMPLOYED-BY *organisation*)
- Information recognised can be:

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes
 - ◇ type information to be extracted must be decided *in advance*
 - ◇ information extracted typically includes mentions of
 - *entities* (e.g. organisations, persons, locations) and
 - *relations* (e.g. *person* IS-EMPLOYED-BY *organisation*)
- Information recognised can be:
 - ◇ extracted and stored in a structured record, e.g. database system

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes
 - ◇ type information to be extracted must be decided *in advance*
 - ◇ information extracted typically includes mentions of
 - *entities* (e.g. organisations, persons, locations) and
 - *relations* (e.g. *person* IS-EMPLOYED-BY *organisation*)
- Information recognised can be:
 - ◇ extracted and stored in a structured record, e.g. database system
 - sometimes called “knowledge base population”

- IR contrasts with *Information Extraction* (IE).
- IE recognises *specific* information in documents, making it available to subsequent automated processes
 - ◇ type information to be extracted must be decided *in advance*
 - ◇ information extracted typically includes mentions of
 - *entities* (e.g. organisations, persons, locations) and
 - *relations* (e.g. *person* IS-EMPLOYED-BY *organisation*)
- Information recognised can be:
 - ◇ extracted and stored in a structured record, e.g. database system
 - sometimes called “knowledge base population”
 - ◇ stored in document itself as embedded mark-up

Applications: Text Categorization

- Task: automatically assign texts to different categories

Applications: Text Categorization

- Task: automatically assign texts to different **categories**
 - ◊ e.g. for email — assign to categories: *junk* vs. *non-junk*

Applications: Text Categorization

- Task: automatically assign texts to different **categories**
 - ◇ e.g. for email — assign to categories: *junk* vs. *non-junk*
 - ◇ e.g. for newspaper articles — assign to categories:
sport vs. *politics* vs. *other*

Applications: Text Categorization

- Task: automatically assign texts to different **categories**
 - ◇ e.g. for email — assign to categories: *junk* vs. *non-junk*
 - ◇ e.g. for newspaper articles — assign to categories: *sport* vs. *politics* vs. *other*
 - ◇ e.g. for product reviews — assign to sentiment categories: *positive* vs. *negative* vs. *neutral*

Applications: Text Categorization

- Task: automatically assign texts to different **categories**
 - ◇ e.g. for email — assign to categories: *junk* vs. *non-junk*
 - ◇ e.g. for newspaper articles — assign to categories: *sport* vs. *politics* vs. *other*
 - ◇ e.g. for product reviews — assign to sentiment categories: *positive* vs. *negative* vs. *neutral*
- Usually achieved by having a set of documents that are representative of each category

Applications: Text Categorization

- Task: automatically assign texts to different **categories**
 - ◇ e.g. for email — assign to categories: *junk* vs. *non-junk*
 - ◇ e.g. for newspaper articles — assign to categories: *sport* vs. *politics* vs. *other*
 - ◇ e.g. for product reviews — assign to sentiment categories: *positive* vs. *negative* vs. *neutral*
- Usually achieved by having a set of documents that are representative of each category
 - ◇ use statistical/probabilistic/neural computational methods to learn a model that predicts which set of documents a new, previously unseen document is most like

Single Document

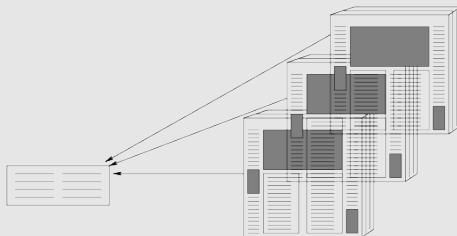


Applications: Summarization

Single Document



Multiple Document



Applications: Natural Language Generation

- Generate natural language text from an abstract, structured representation. For example:

Applications: Natural Language Generation

- Generate natural language text from an abstract, structured representation. For example:
 - ◇ generate an equipment maintenance manual in multiple languages from a single abstract representation of steps to be carried out

Applications: Natural Language Generation

- Generate natural language text from an abstract, structured representation. For example:
 - ◇ generate an equipment maintenance manual in multiple languages from a single abstract representation of steps to be carried out
 - ◇ generate a readable summary of several hours of numerical output from sensors on a patient in intensive care

Applications: Natural Language Generation

- Generate natural language text from an abstract, structured representation. For example:
 - ◇ generate an equipment maintenance manual in multiple languages from a single abstract representation of steps to be carried out
 - ◇ generate a readable summary of several hours of numerical output from sensors on a patient in intensive care
 - ◇ generate content for a personalised spoken tour for an individual visiting a museum given the individual's interests and stored, abstract representations describing items held in the museum

Applications: Natural Language Generation

- Generate natural language text from an abstract, structured representation. For example:
 - ◇ generate an equipment maintenance manual in multiple languages from a single abstract representation of steps to be carried out
 - ◇ generate a readable summary of several hours of numerical output from sensors on a patient in intensive care
 - ◇ generate content for a personalised spoken tour for an individual visiting a museum given the individual's interests and stored, abstract representations describing items held in the museum
- Doing this well involves moving beyond printing canned text in a pre-specified format

Applications: Natural Language Generation

- Generate natural language text from an abstract, structured representation. For example:
 - ◇ generate an equipment maintenance manual in multiple languages from a single abstract representation of steps to be carried out
 - ◇ generate a readable summary of several hours of numerical output from sensors on a patient in intensive care
 - ◇ generate content for a personalised spoken tour for an individual visiting a museum given the individual's interests and stored, abstract representations describing items held in the museum
- Doing this well involves moving beyond printing canned text in a pre-specified format
- Need to make choices about: selection and ordering of content, which words/syntactic structures to use, when to introduce anaphors (e.g. pronouns), etc.

Applications: Machine Translation

- Translate text from one language to another



Applications: Machine Translation

- Translate text from one language to another
e.g. English to French and/or vice versa



Applications: Machine Translation

- Translate text from one language to another
e.g. English to French and/or vice versa
- One solution: write a computer program to do the translation.



Applications: Machine Translation

- Translate text from one language to another
e.g. English to French and/or vice versa
- One solution: write a computer program to do the translation.
- **Very challenging problem!**



Applications: Machine Translation

- Translate text from one language to another
e.g. English to French and/or vice versa
- One solution: write a computer program to do the translation.
- Very challenging problem!
- Requires immense amount of knowledge about language and the world.



Applications: Machine Translation

- Translate text from one language to another
e.g. English to French and/or vice versa
- One solution: write a computer program to do the translation.
- **Very challenging problem!**
- Requires immense amount of knowledge about language and the world.
- **Better solution: Learn from corpora that are translations of each other.**



Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◊ Yes – of course!

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◊ Yes – of course!
- Human decisions may be based on incorrect or limited information derived from imperfect information access tools

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◊ Yes – of course!
- **Human decisions** may be based on incorrect or limited information derived from imperfect information access tools
 - ◊ **search engines/text mining tools may fail to retrieve relevant info**

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◇ Yes – of course!
- **Human decisions** may be based on incorrect or limited information derived from imperfect information access tools
 - ◇ search engines/text mining tools may fail to retrieve relevant info
 - ◇ **machine translation tools may mistranslate key passages**

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◇ Yes – of course!
- **Human decisions** may be based on incorrect or limited information derived from imperfect information access tools
 - ◇ search engines/text mining tools may fail to retrieve relevant info
 - ◇ machine translation tools may mistranslate key passages
- **Profiles of individuals may be gathered/inferred from the language they use (under GDPR language is personal data)**

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◊ Yes – of course!
- **Human decisions** may be based on incorrect or limited information derived from imperfect information access tools
 - ◊ search engines/text mining tools may fail to retrieve relevant info
 - ◊ machine translation tools may mistranslate key passages
- **Profiles of individuals** may be gathered/inferred from the language they use (under GDPR language is personal data)
 - ◊ **opinions expressed on-line**

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◇ Yes – of course!
- **Human decisions** may be based on incorrect or limited information derived from imperfect information access tools
 - ◇ search engines/text mining tools may fail to retrieve relevant info
 - ◇ machine translation tools may mistranslate key passages
- **Profiles of individuals** may be gathered/inferred from the language they use (under GDPR language is personal data)
 - ◇ opinions expressed on-line
 - ◇ **decisions made by gender/age/IQ/race/sexual orientation classifiers**

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◊ Yes – of course!
- **Human decisions** may be based on incorrect or limited information derived from imperfect information access tools
 - ◊ search engines/text mining tools may fail to retrieve relevant info
 - ◊ machine translation tools may mistranslate key passages
- **Profiles of individuals** may be gathered/inferred from the language they use (under GDPR language is personal data)
 - ◊ opinions expressed on-line
 - ◊ decisions made by gender/age/IQ/race/sexual orientation classifiers
- See: **UKRI Framework for responsible innovation (AREA: anticipate, reflect, engage, act)**

Text Processing/NLP: Ethical and Social Considerations

- Are there ethical and social implications for research and innovation in text processing and NLP?
 - ◊ Yes – of course!
- **Human decisions** may be based on incorrect or limited information derived from imperfect information access tools
 - ◊ search engines/text mining tools may fail to retrieve relevant info
 - ◊ machine translation tools may mistranslate key passages
- **Profiles of individuals** may be gathered/inferred from the language they use (under GDPR language is personal data)
 - ◊ opinions expressed on-line
 - ◊ decisions made by gender/age/IQ/race/sexual orientation classifiers
- See: UKRI Framework for responsible innovation (AREA: anticipate, reflect, engage, act)
- See: **ACL Ethics for NLP resources:**
https://aclweb.org/aclwiki/Ethics_in_NLP

- Text Encoding

}

Rob

}

Carol

Course Outline

- Text Encoding
- Text Compression

}

Rob

}

Carol

Course Outline

- Text Encoding
- Text Compression
- Information Retrieval



Rob



Carol

Course Outline

- Text Encoding
- Text Compression
- Information Retrieval
- Sentiment Analysis



Rob



Carol

Course Outline

- Text Encoding
- Text Compression
- Information Retrieval
- Sentiment Analysis
- Information Extraction



Rob



Carol

Course Outline

- Text Encoding
- Text Compression
- Information Retrieval
- Sentiment Analysis
- Information Extraction
- Introduction to Deep Learning for Text Processing



Rob



Carol

- Two hours each week starting in week 1
- Exercises designed to let you learn Python techniques for text processing
- Helpful for doing the assignments
- Opportunities to ask questions about the assignments

- Two assignments ($2 \times 25\%$)
 - ◇ Information Retrieval: out week 5; in week 8
 - ◇ Sentiment Analysis: out week 9; in week 12
- One exam (50%)
 - ◇ During the exam period in January
- COM4115 (15 credits) distinguished from COM3110 (10 credits) by more challenging/extended assignments.

Major sources:

- Programming — see module homepage for suggestions
- Information Retrieval
 - ◇ Baeza-Yates and Ribeiro-Neto, Modern Information Retrieval. New York: ACM Press, 1999.
 - ◇ C. Manning, P. Raghavan and H. Schtze, Introduction to Information Retrieval, Cambridge University Press. 2008.
- General:
 - ◇ C.D. Manning and H. Schütze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.
 - ◇ D. Jurafsky and J. Martin, Speech and Language Processing, Prentice-Hall, 2007 (2nd edn) (incomplete 3rd edition available free on the web).